



Aricept®EVESS 5mg, 10mg

Donepezil Hydrochloride Orodispersible Tablet

1. **TRADE NAME OF THE MEDICINAL PRODUCT**

Aricept Evess 5 mg orodispersible tablet
Aricept Evess 10 mg orodispersible tablet

2. **QUALITATIVE AND QUANTITATIVE COMPOSITION**

5 mg donepezil hydrochloride tablets each containing 4.56 mg donepezil free base
10 mg donepezil hydrochloride tablets each containing 9.12 mg donepezil free base
For excipients, see 6.1

3. **PHARMACEUTICAL FORM**

Orodispersible Tablet
White tablet debossed with "5" on one side and "Aricept" on the other side.
Yellow tablet debossed with "10" on one side and "Aricept" on the other side.

4. **CLINICAL PARTICULARS**

4.1 **Therapeutic indication**

Aricept Evess tablets are indicated for the treatment of mild, moderate, and severe dementia in Alzheimer's Disease.

4.2 **Posology and method of administration**

Adults/Elderly:

Mild to Moderate Alzheimer's Disease

Treatment is initiated at 5mg/day (once-a-day dosing). Aricept Evess should be taken orally, in the evening, just prior to retiring. The tablet should be placed on the tongue and allowed to disintegrate before swallowing with or without water, according to patient preference. The 5mg/day dose should be maintained for at least one month in order to allow the earliest clinical responses to treatment to be assessed and to allow steady-state concentrations of donepezil hydrochloride to be achieved. Following a one-month clinical assessment of treatment at 5mg/day, the dose of Aricept Evess can be increased to 10mg/day (once-a-day dosing). The maximum recommended daily dose is 10mg. Doses greater than 10mg/day have not been studied in clinical trials.

Upon discontinuation of treatment, a gradual abatement of the beneficial effects of Aricept is seen.

Severe Alzheimer's Disease

Aricept has been shown to be effective in controlled clinical trials at a dose of 10mg administered once daily.

Evidence from the controlled trials in mild to moderate Alzheimer's Disease indicates that the 10mg dose, with one week titration, is likely to be associated with a higher incidence of cholinergic adverse events than the 5mg dose. In open label trials using a 6 week titration, the frequency of these same adverse events was similar between the 5mg and 10mg dose groups. Therefore, because steady state is not achieved for 15 days and because the incidence of untoward effects may be influenced by the rate of dose escalation, a dose of 10mg shouldnot be escalated until patients have been on a daily dose of 5mg for 4 to 6 weeks. Upon discontinuation of treatment, a gradual abatement of the beneficial effects of Aricept is seen. There is no evidence of a rebound effect after abrupt discontinuation of therapy.

Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia. Diagnosis should be made according to accepted guidelines (e.g. DSM IV, ICD 10). Therapy with donepezil should only be started if a caregiver is available who will regularly monitor drug intake for the patient. Maintenance treatment can be continued for as long as a therapeutic benefit for the patient exists. Therefore, the clinical benefit of donepezil should be reassessed on a regular basis. Discontinuation should be considered when evidence of a therapeutic *effect* is no longer present. Individual response to donepezil cannot be predicted.

Renal and hepatic impairment:

A similar dose schedule can be followed for patients with renal impairment as clearance of donepezil hydrochloride is not affected by this condition. Due to possible increased exposure in mild to moderate hepatic impairment (see section 5.2), dose escalation should be performed according to individual tolerability. There are no data for patients with severe hepatic impairment.

Children:

ARICEPT is not recommended for use in children.

4.3 **Contraindications**

ARICEPT is contraindicated in patients with a known hypersensitivity to donepezil hydrochloride, piperidine derivatives, or to any excipients used in the formulation.

4.4 **Special warnings and special precautions for use**

The use of Aricept in patients with severe dementia, other types of dementia or other types of memory impairment (e.g., age-related cognitive decline), has not been investigated.

Anaesthesia: Aricept, as a cholinesterase inhibitor, is likely to exaggerate succinylcholine-type muscle relaxation during anaesthesia.

Cardiovascular Conditions: Because of their pharmacological action, cholinesterase inhibitors may have vagotonic effects on heart rate (e.g. bradycardia). The potential for this action may be particularly important to patients with "sick sinus syndrome" or other supraventricular cardiac conduction conditions, such as sinoatrial or atrioventricular block.

There have been reports of syncope and seizures. In investigating such patients the possibility of heart block or long sinus pauses should be considered.

There have been post-marketing reports of QTc interval prolongation and Torsade de Pointes (see sections 4.5 and 4.8). Caution is advised in patients with pre-existing or family history of QTc prolongation, in patients treated with drugs affecting the QTc interval, or in patients with relevant pre-existing cardiac disease (e.g. uncompensated heart failure, recent myocardial infarction, bradyarrhythmias), or electrolyte disturbances (hypokalaemia, hypomagnesaemia). Clinical monitoring (ECG) may be required.

Gastrointestinal Conditions: Patients at increased risk for developing ulcers, e.g., those with a history of ulcer disease or those receiving concurrent nonsteroidal anti-inflammatory drugs (NSAIDs), should be monitored for symptoms. However, the clinical studies with Aricept showed no increase, relative to placebo, in the incidence of either peptic ulcer disease or gastrointestinal bleeding.

Genitourinary: Although not observed in clinical trials of Aricept, cholinomimetics may cause bladder outflow obstruction.

Neurological Conditions: Seizures: Cholinomimetics are believed to have some potential to cause generalised convulsions. However, seizure activity may also be a manifestation of Alzheimer's disease.

Cholinomimetics may have the potential to exacerbate or induce extrapyramidal symptoms.

Neuroleptic Malignant Syndrome (NMS): NMS, a potentially life-threatening condition characterised by hyperthermia, muscle rigidity, autonomic instability, altered consciousness and elevated serum creatine phosphokinase levels, has been reported to occur very rarely in association with donepezil, particularly in patients also receiving concomitant antipsychotics. Additional signs may include myoglobinuria (rhabdomyolysis) and acute renal failure. If a patient

develops signs and symptoms indicative of NMS, or presents with unexplained high fever without additional clinical manifestations of NMS, treatment should be discontinued.

Pulmonary Conditions: Because of their cholinomimetic actions, cholinesterase inhibitors should be prescribed with care to patients with a history of asthma or obstructive pulmonary disease.

The administration of Aricept concomitantly with other inhibitors of acetylcholinesterase, agonists or antagonists of the cholinergic system should be avoided.

Severe Hepatic Impairment: There are no data for patients with severe hepatic impairment.

Mortality in Vascular Dementia Clinical Trials:

Three clinical trials of 6 months duration were conducted studying individuals meeting the NINDS-AIREN criteria for probable or possible vascular dementia (VaD). The NINDS-AIREN criteria are designed to identify patients whose dementia appears to be due solely to vascular causes and to exclude patients with Alzheimer's disease. In the first study, the mortality rates were 2/198 (1.0%) on donepezil hydrochloride 5 mg, 5/206 (2.4%) on donepezil hydrochloride 10 mg and 7/199 (3.5%) on placebo. In the second study, the mortality rates were 4/208 (1.9%) on donepezil hydrochloride 5 mg, 3/215 (1.4%) on donepezil hydrochloride 10 mg and 1/193 (0.5%) on placebo. In the third study, the mortality rates were 11/648 (1.7%) on donepezil hydrochloride 5 mg and 0/326 (0%) on placebo. The mortality rate for the three VaD studies combined in the donepezil hydrochloride group (1.7%) was numerically higher than in the placebo group (1.1%), however, this difference was not statistically significant.

The majority of deaths in patients taking either donepezil hydrochloride or placebo appear to result from various vascular related causes, which could be expected in this elderly population with underlying vascular disease. An analysis of all serious non-fatal and fatal vascular events showed no difference in the rate of occurrence in the donepezil hydrochloride group relative to placebo.

In pooled Alzheimer's disease studies (n=4146), and when these Alzheimer's disease studies were pooled with other dementia studies including the vascular dementia studies (total n=6888), the mortality rate in the placebo groups numerically exceeded that in the donepezil hydrochloride groups.

4.5 **Interaction with other medicaments and other forms of interaction**

Donepezil hydrochloride and/or any of its metabolites does not inhibit the metabolism of theophylline, warfarin, cimetidine or digoxin in humans. The metabolism of donepezil hydrochloride is not affected by concurrent administration of digoxin or cimetidine.

In vitro studies have shown that the cytochrome P450 isoenzymes 3A4 and to a minor extent 2D6 are involved in the metabolism of donepezil. Drug interaction studies performed *in vitro* show that ketoconazole and quinidine, inhibitors of CYP3A4 and 2D6 respectively, inhibit donepezil metabolism. Therefore, these and other CYP3A4 inhibitors, such as itraconazole and erythromycin, and CYP2D6 inhibitors, such as fluoxetine, could inhibit the metabolism of donepezil. In a study in healthy volunteers, ketoconazole increased mean donepezil concentrations by about 30%. Enzyme inducers, such as rifampicin, phenytoin, carbamazepine and alcohol may reduce the levels of donepezil. Since the magnitude of an inhibiting or inducing *effect* is unknown, such drug combinations should be used with care. Donepezil hydrochloride has the potential to interfere with medications having anticholinergic activity. There is also the potential for synergistic activity with concomitant treatment involving medications such as succinylcholine, other neuro-muscular blocking agents or cholinergic agonists or beta blocking agents that have *effects* on cardiac conduction. Cases of QTc interval prolongation and Torsade de Pointes have been reported for donepezil. Caution is advised when donepezil is used in combination with other medicinal products known to prolong the QTc interval and clinical monitoring (ECG) may be required. Examples include:

- Class IA antiarrhythmics (e.g. quinidine)
- Class III antiarrhythmics (e.g. amiodarone, sotalol)
- Certain antidepressants (e.g. citalopram, escitalopram, amitriptyline)
- Other antipsychotics (e.g. phenothiazine derivatives, sertindole, pimozide, ziprasidone)
- Certain antibiotics (e.g. clarithromycin, erythromycin, levofloxacin, moxifloxacin)

4.6 **Pregnancy and lactation**

Pregnancy:

There are no adequate data from the use of donepezil in pregnant women. Studies in animals have not shown teratogenic *effect* but have shown pre and post natal toxicity (see section 5.3 preclinical safety data). The potential risk for humans is unknown.

Aricept should not be used during pregnancy unless clearly necessary.

Lactation:

Donepezil is excreted in the milk of rat. It is not known whether donepezil hydrochloride is excreted in human breast milk and there are no studies in lactating women. Therefore, women on donepezil should not breast feed.

4.7 **Effects on ability to drive and use machines**

Donepezil has minor or moderate influence on the ability to drive and use machines. Dementia may cause impairment of driving performance or compromise the ability to use machinery. Furthermore, donepezil can induce fatigue, dizziness and muscle cramps, mainly when initiating or increasing the dose. The treating physician should routinely evaluate the ability of patients on donepezil to continue driving or operating complex machines.

4.8 **Undesirable effects**

The most common adverse events are diarrhea, muscle cramps, fatigue, nausea, vomiting and insomnia.

Adverse reactions reported as more than an isolated case are listed below, by system organ class and by frequency. Frequencies are defined as: very common (≥ 1/10), common (≥ 1/100 to < 1/10), uncommon (≥ 1/1,000 to 1/100), rare (≥ 1/10,000 to 1/1,000); very rare (< 1/10,000) and not known (cannot be estimated from available data).

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare	Not known
Infections and infestation s		Common cold				
Metabolism and nutrition disorders		Anorexia				
Psychiatric disorders		Hallucinations** Agitation** Aggressive behaviour** Abnormal dreams and Nightmares**				
Nervous system disorders		Syncope* Dizziness Insomnia	Seizure*	Extrapyramidal symptoms	Neurolepti c malignant syndrome	

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare	Not known
Cardiac disorders			Bradycardia	Sino-atrial block Atrioventricular block		Polymorphic ventricular tachycardia including Torsade de Pointes Electrocardiogram QT interval prolonged
Gastrointestinal disorders	Diarrhoea Nausea	Vomiting Abdominal disturbance	Gastrointestinal haemorrhage Gastric and duodenal ulcers Salivary hypersecretion			
Hepato-biliary disorders				Liver dysfunction including hepatitis***		
Skin and subcutaneous tissue disorders		Rash Pruritis				
Musculo-skeletal, connective tissue and bone disorders		Muscle cramps				
Renal and urinary disorders		Urinary incontinence				
General disorders and administration site conditions	Headache	Fatigue Pain				
Investigations			Minor increase in serum concentration of muscle creatine kinase			
Injury and poisoning		Accidents including falls				

*In investigating patients for syncope or seizure the possibility of heart block or long sinus pauses should be considered (see section 4.4)

*Reports of hallucinations, abnormal dreams, nightmares, agitation and aggressive behaviour have resolved on dose-reduction or discontinuation of treatment.

***In cases of unexplained liver dysfunction, withdrawal of Aricept Evess should be considered.

Post-Marketing Experience

There have been post-marketing reports of hallucinations, agitation, aggressive behavior, seizure, hepatitis, gastric ulcer, duodenal ulcer, and gastrointestinal hemorrhage.

4.9 Overdose

The estimated median lethal dose of donepezil hydrochloride following administration of a single oral dose in mice and rats in 45 and 32 mg/kg, respectively, or approximately 225 and 160 times the maximum recommended human dose of 10 mg per day. Dose-related signs of cholinergic stimulation were observed in animals and included reduced spontaneous movement, prone position, staggering gait, lacrimation, clonic convulsions, depressed respiration, salivation, miosis, fasciculation and lower body surface temperature.

Overdosage with cholinesterase inhibitors can result in cholinergic crisis characterized by severe nausea, vomiting, salivation, sweating, bradycardia, hypotension, respiratory depression, collapse and convulsions. Increasing muscle weakness is a possibility and may result in death if respiratory muscles are involved.

As in any case of overdose, general supportive measures should be utilised. Tertiary anticholinergics such as atropine may be used as an antidote for Aricept overdosage. Intravenous atropine sulphate titrated to effect is recommended: an initial dose of 1.0 to 2.0 mg IV with subsequent doses based upon clinical response. Atypical responses in blood pressure and heart rate have been reported with other cholinomimetics when co-administered with quaternary anticholinergics such as glycopyrrolate. It is not known whether donepezil hydrochloride and/or its metabolites can be removed by dialysis (hemodialysis, peritoneal dialysis, or hemofiltration).

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Donepezil hydrochloride is a specific and reversible inhibitor of acetylcholinesterase, the predominant cholinesterase in the brain. Donepezil hydrochloride is *in vitro* over 1000 times more potent an inhibitor of this enzyme than of butyrylcholinesterase, an enzyme which is present mainly outside the central nervous system.

Alzheimer's Dementia

In patients with Alzheimer's dementia participating in clinical trials, administration of single daily doses of 5 mg or 10 mg of Aricept produced steady-state inhibition of acetylcholinesterase activity (measured in erythrocyte membranes) of 63.6% and 77.3%, respectively when measured post dose. The inhibition of acetylcholinesterase (AChE) in red blood cells by donepezil hydrochloride has been shown to correlate to changes in ADAS-cog, a sensitive scale which examines selected aspects of cognition. The potential for donepezil hydrochloride to alter the course of the underlying neuropathology has not been studied. Thus, Aricept cannot be considered to have any effect on the progress of the disease.

Efficacy of treatment of Alzheimer's Dementia with Aricept has been investigated in four placebo-controlled trials, 2 trials of 6-month duration and 2 trials of 1-year duration.

In the 6 months clinical trial, an analysis was done at the conclusion of donepezil treatment using a combination of three efficacy criteria: the ADAS-Cog (a measure of cognitive performance), the Clinician Interview Based Impression of Change with Caregiver Input (a measure of global function) and the Activities of Daily Living Subscale of the Clinical Dementia Rating Scale (a measure of capabilities in community affairs, home and hobbies and personal care).

Patients who fulfilled the criteria listed below were considered treatment responders.

Response = Improvement of ADAS-Cog of at least 4 points

No deterioration of CIBIC +

No Deterioration of Activities of Daily Living Subscale of the Clinical Dementia Rating Scale

	% Response Intent to Treat Population n = 365	Evaluable Population n = 352
Placebo Group	10%	10%
Aricept 5-mg Group	18%*	18%*
Aricept 10-mg Group	21%*	22%**

*p < 0.05

**p < 0.01

Aricept produced a dose-dependent statistically significant increase in the percentage of patients who were judged treatment responders.

5.2 Pharmacokinetic properties

Absorption: Maximum plasma levels are reached approximately 3 to 4 hours after oral administration. Plasma concentrations and area under the curve rise in proportion to the dose. The terminal disposition half-life is approximately 70 hours, thus, administration of multiple single-daily doses results in gradual approach to steady-state. Approximate steady-state is achieved within 3 weeks after initiation of therapy. Once at steady-state, plasma donepezil hydrochloride concentrations and the related pharmacodynamic activity show little variability over the course of the day.

Food did not affect the absorption of donepezil hydrochloride.

Distribution: Donepezil hydrochloride is approximately 95% bound to human plasma proteins. The plasma protein binding of the active metabolite 6-O-desmethyl donepezil is not known. The distribution of donepezil hydrochloride in various body tissues has not been definitively studied. However, in a mass balance study conducted in healthy male volunteers, 240 hours after the administration of a single 5 mg dose of ¹⁴C-labeled donepezil hydrochloride, approximately 28% of the label remained unrecovered. This suggests that donepezil hydrochloride and/or its metabolites may persist in the body for more than 10 days.

Metabolism/Excretion: Donepezil hydrochloride is both excreted in the urine intact and metabolised by the cytochrome P450 system to multiple metabolites, not all of which have been identified. Following administration of a single 5 mg dose of ¹⁴C-labeled donepezil hydrochloride, plasma radioactivity, expressed as a percent of the administered dose, was present primarily as intact donepezil hydrochloride (30%), 6-O-desmethyl donepezil (11% - only metabolite that exhibits activity similar to donepezil hydrochloride), donepezil-cis-N-oxide (9%), 5-O-desmethyl donepezil (7%) and the glucuronide conjugate of 5-O-desmethyl donepezil (3%).

Approximately 57% of the total administered radioactivity was recovered from the urine (17% as unchanged donepezil), and 14.5% was recovered from the faeces, suggesting biotransformation and urinary excretion as the primary routes of elimination. There is no evidence to suggest enterohepatic recirculation of donepezil hydrochloride and/or any of its metabolites.

Plasma donepezil concentrations decline with a half-life of approximately 70 hours.

Sex, race and smoking history have no clinically significant influence on plasma concentrations of donepezil hydrochloride. The pharmacokinetics of donepezil has not been formally studied in healthy elderly subjects, or in Alzheimer's or vascular dementia patients. However, mean plasma levels in patients closely agreed with those of young healthy volunteers.

Patients with mild to moderate hepatic impairment had increased donepezil steady state concentrations; mean AUC by 48% and mean Cmax by 39% (see section 4.2).

5.3 Preclinical safety data

Extensive testing in experimental animals has demonstrated that this compound causes few effects other than the intended pharmacological effects consistent with its action as a cholinergic stimulator (See section 4.9).

Donepezil is not mutagenic in Ames reverse mutation assay in bacteria or in the mouse lymphoma forward mutation assay *in vitro*. Donepezil hydrochloride did not induce unscheduled DNA synthesis in rat primary hepatocyte cultures following oral dosing of the animals.

Some clastogenic effects were observed in Chinese Hamster Lung cells *in vitro*, but only at concentrations approaching those overtly toxic to the cells and more than 3000 times the steady-state plasma concentrations observed in patients during clinical trials. Donepezil hydrochloride was not clastogenic in the mouse micronucleus model *in vivo*.

Donepezil hydrochloride had no effect on fertility in rats and was not teratogenic in rats or rabbits, but had a slight effect on stillbirths and early pup survival when administered to pregnant rats at 50 times the human dose (see Section 4.6).

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Mannitol
Silica, colloidal anhydrous
k-carrageenan
Polyvinyl alcohol
Additionally, the 10mg tablet contains ferric oxide (yellow)

6.2 Incompatibilities

None

6.3 Shelf life

36 months

6.4 Special precautions for storage

Do not store above 30°C.

Tablet may change color with light, so keep in aluminium pouch until taken.

6.5 Nature and contents of container

5 mg tablets:, 10 mg tablets:

Unit dose blister strips of 28 tablets box

6.6 Instruction for use/handling

No special instructions.

Manufactured by

Bushu Pharmaceuticals Ltd. Misato Factory
950, Hiraki, Ohaza, Misato-machi,
Kodama-gun, Saitama-ken, Japan

Repacked by

Interthai Pharmaceutical Manufacturing Ltd.
1899 Phaholyothin Road, Ladyao, Chatuchak,
Bangkok 10900, Thailand.

Product Registration Holder

Eisai (Malaysia) Sdn. Bhd.
Unit 701D, Level 7, Tower D, Uptown 5
No. 5, Jalan SS21/39, Damansara Uptown
47400 Petaling Jaya
Selangor, Malaysia

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